Spatiotemporal Signal Propagation
3-Hour Classroom Worksheet

Systems Biology — University of Warsaw

Prerequisites: *Spatiotemporal Signal Propagation — Mathematical Formalization* (companion handout). Notation is identical throughout.

Scripts: `scripts/spatiotemporal\_signal\_propagation.py`, `scripts/compare\_spatiotemporal\_behavior.py`

Session overview.

Part I (90 min) Guided exploration of the analysis pipeline. You will run single-block and cohort-level analyses, interpret outputs, and develop intuition for the effect of parameter choices.

Part II (90 min) Critical analysis and algorithmic improvement. You will identify three concrete mathematical limitations of the current approach, derive improved estimators on paper, and implement them.

Part I Guided Exploration (90 min)

Block 1 (25 min) Your first single-block run

Task 1 (Setup and first run)**.** Navigate to the project root and run the single-block analysis on experiment 1, site 1 (a wild-type condition).

```
python scripts/spatiotemporal_signal_propagation.py \
    --exp-id 1 --site-id 1 \
    --signal-col ERKKTR_ratio \
    --spatial-radius 60 \
    --future-window-frames 3 \
    --jump-quantile 0.9 \
    --output-dir analysis_outputs
```

The script prints a short report. Locate each printed quantity in the mathematical formalism of the companion handout.

(a) How many nodes $|V|$ and spatial edges $|E_{\text{sp}}|$ were created? What is the ratio $|E_{\text{sp}}|/|V|$? What does this ratio tell you about the local density of the cell monolayer?
(b) What is the data-driven threshold $\theta_{0.90}$? Using Definition 4 from the handout, write out in words what this value means for a cell at an arbitrary node (k, t).
(c) Record p_{exp}, p_{unexp}, RD, and RR. Is the relative risk RR greater than, equal to, or less than 1? State in one biological sentence what this implies.

Task 2 (Inspect the output tables)**.** Open the following file in a spreadsheet or with Python (`pd.read_csv(..., compression='gzip')`):

```
analysis_outputs/exp_1_site_1_ERKKTR_ratio/nodes.csv.gz
```

(a) For three tracks of your choice, plot $S_k(t)$ over time (the column `signal_value`). Mark the frames where $\mathcal{J}_k(t) = 1$ (`jump_event` = True) with a vertical line or a dot. Are the jumps isolated single frames, or do they appear in bursts?
(b) Filter to rows where `future_self_jump` is True *and* `neighbor_jump_now` is True. How many such rows are there? Express this count as a fraction of all `neighbor_jump_now` = True rows. Verify that this matches p_{exp} from Task 1.
(c) Compute the mean and standard deviation of `neighbor_count` across all nodes. Sketch (by hand or code) a histogram of `neighbor_count`. At radius $r = 60$, is the neighbourhood sparse or dense?

Block 2 (30 min) Parameter sensitivity

Keeping the same block (exp 1, site 1, ERK signal), re-run the script with the following parameter variations. Fill in the table below each time; record at minimum θ, $|E_{\text{sp}}|$, p_{exp}, p_{unexp}, RR.

Task 3 (Varying the spatial radius r)**.** Run with $r \in \{30, 60, 90, 150\}$.

r	$\lvert E_{\mathrm{sp}}\rvert$	p_{exp}	p_{unexp}	RR
30				
60				
90				
150				

(a) As r grows, $|E_{\mathrm{sp}}|$ grows roughly how fast? (Hint: for a uniform cell density ρ [cells per unit area], the expected number of neighbours within radius r is $\mathbb{E}[n(k,t)] \approx \pi r^2 \rho$. Does $|E_{\mathrm{sp}}|$ scale as r^2?)
(b) Does RR increase or decrease with r? Explain why a very large r should push RR toward 1. (Hint: consider what happens when almost every cell is a neighbour of every other cell.)

Task 4 (Varying the future window W)**.** Run with $W \in \{1,\ 3,\ 6,\ 12\}$ (keeping $r = 60$).

W (frames)	$W \cdot 5$ min	p_{exp}	p_{unexp}	RR
1	5 min			
3	15 min			
6	30 min			
12	60 min			

(a) Does p_{exp} increase or decrease with W? Is the same trend observed for p_{unexp}? Why does both rates rising not necessarily imply that RR changes?
(b) At what window length does RR peak? What does this suggest about the timescale of cell-to-cell ERK coordination in this dataset?

Task 5 (Varying the jump quantile q)**.** Run with $q \in \{0.70,\ 0.80,\ 0.90,\ 0.95\}$ (keeping $r = 60$, $W = 3$).

q	θ_q	p_{exp}	p_{unexp}	RR
0.70				
0.80				
0.90				
0.95				

(a) As q increases, fewer nodes satisfy $\mathcal{J}_k(t) = 1$. How does the fraction of jump events scale with q?
(b) Does increasing q (i.e. using a stricter threshold for calling a "jump") increase or decrease RR? Propose a biological interpretation: are rarer, larger jumps more or less spatially coordinated in this dataset?

Block 3 (35 min) Comparing individual conditions and running a cohort

Task 6 (Manual pairwise comparison)**.** According to the metadata file

`01-readme-experiment-description_2022-04-05.csv`

sites 1–4 are WT and sites 9–12 are PIK3CA E545K. Run the single-block script separately for **site 1 (WT)** and **site 9 (PIK3CA E545K)**.

Fill in the table (use the same $r = 60$, $W = 3$, $q = 0.90$):

Condition	Site	θ	p_{exp}	p_{unexp}	RR
WT	1				
PIK3CA E545K	9				

(a) Is RR higher or lower for the PIK3CA mutant compared to WT? PIK3CA E545K is a gain-of-function oncogenic mutation that hyperactivates PI3K signalling. Based on your result, does hyperactive PI3K increase or decrease ERK-activity coordination among neighbours?
(b) Can you directly compare the two θ values? Why or why not? (This question anticipates Limitation L2 in Part II.)

Task 7 (Cohort-level analysis with the comparison script)**.** Run the multi-block comparison script to aggregate all available blocks and compare by mutation:

```
python scripts/compare_spatiotemporal_behavior.py \
    --signal-col ERKKTR_ratio \
    --group-by mutation \
    --spatial-radius 60 \
    --future-window-frames 3 \
    --jump-quantile 0.9 \
    --exclude-mutations ErbB2 \
    --output-dir analysis_outputs
```

Open the group-level summary:

`analysis_outputs/comparison_mutation_ERKKTR_ratio/group_level_summary.csv`

(a) Which mutation has the *highest* $\overline{\text{RR}}$? Which has the lowest? Does the ordering match your expectation from the PI3K pathway hierarchy (PI3K $\to$ AKT $\to$ PTEN)?
(b) Note the column `n_blocks` for each group. Some mutations have more replicate blocks than others. Why is $\overline{\text{RR}}$ (the mean over blocks) a fragile summary when `n_blocks` is small?
(c) Repeat the cohort run with `-signal-col FoxO3A_ratio`. Fill in the comparison table:

Mutation	$\overline{\text{RR}}$ (ERK)	$\overline{\text{RR}}$ (FoxO3A)
WT		
AKT1 E17K		
PIK3CA E545K		
PIK3CA H1047R		
PTEN del		

Is ERK or FoxO3A more spatially coordinated overall? Given that FoxO3A is a direct AKT substrate while ERK is a MAPK downstream of RAS, does the difference make biological sense?

Part II Critical Analysis and Algorithmic Improvement (90 min)

The analysis you have run so far is a solid first pass. In this part you will identify three mathematical limitations, see concretely why each one matters, and then derive and implement an improved estimator. All improvements are achievable with NumPy/pandas without modifying the original scripts beyond a new analysis function.

Limitation L1 (25 min) Simultaneity: exposure and response are not properly ordered in time

The problem

Recall the exposure indicator (Definition 5 of the handout):

$$E_k(t) \;=\; \mathbf{1}[m(k,t) > 0] \qquad \text{where} \qquad m(k,t) = \sum_{(k',t)\,\in\,\mathcal{N}_k(t)} \mathcal{J}_{k'}(t).$$

The neighbour's jump $\mathcal{J}_{k'}(t)$ is measured *at the same frame* t as the focal cell is observed. But the future self-jump is defined as $\mathcal{F}_k(t) = \mathbf{1}[\exists\, t' \in (t, t{+}W] : \mathcal{J}_k(t') = 1]$.

The temporal ordering problem. Both the neighbour's jump $\mathcal{J}_{k'}(t)$ and the future-jump window $(t, t{+}W]$ are anchored to the same frame t. If cell k' jumps at frame t and cell k jumps at frame $t+1$, the current analysis classifies this as a "successful propagation event". But it could equally be that both cells respond to the same upstream stimulus arriving at frame t — there is no *temporal precedence* in the measurement. More critically, the current formula also counts cases where *cell k itself jumps simultaneously* with its neighbour (at frame t) and then again within $(t, t+W]$, even if the two events are independent.

Frame	$\mathcal{J}_k(t)$	$\mathcal{J}_{k'}(t)$	$E_k(t)$	$\mathcal{F}_k(t)$ (W=3)
t	0	1	1	?
$t+1$	1	0	0	
$t+2$	0	0	0	
$t+3$	0	0	0	

Question 1. In the table above, what is $\mathcal{F}_k(t)$? Does the sequence truly suggest that cell k' "caused" cell k to jump? Can you construct a scenario where the assignment $\mathcal{F}_k(t) = 1$ is misleading?

Proposed improvement: lagged exposure

Proposed improvement: Lagged exposure $E_k^{(\tau)}(t)$

For a *lag* $\tau \in \{1, 2, \ldots\}$ (in frames), define the **lagged exposure indicator**

$$E_k(t)^{(\tau)} \;=\; \mathbf{1}[m(k,\, t-\tau) > 0]\,,$$

where $m(k, t-\tau)$ counts how many of cell k's spatial neighbours at frame $t-\tau$ were jumping at that earlier frame. The lagged relative risk is then

$$\mathrm{RR}^{(\tau)} \;=\; \frac{\Pr\!\big(\mathcal{F}_k(t) = 1 \mid E_k(t)^{(\tau)} = 1\big)}{\Pr\!\big(\mathcal{F}_k(t) = 1 \mid E_k(t)^{(\tau)} = 0\big)}.$$

For $\tau = 0$ this reduces to the original RR. For $\tau \geq 1$, the neighbour's jump *strictly precedes* the focal cell's look-ahead window, providing genuine temporal precedence.

The key insight is: if signal truly propagates from a jumping neighbour *to* the focal cell, we expect $\mathrm{RR}^{(\tau)} > 1$ for small lags $\tau \approx 1$–3 and then $\mathrm{RR}^{(\tau)} \to 1$ as τ grows beyond the signalling timescale.

Task 8 (Implement and plot $\mathrm{RR}^{(\tau)}$ vs. τ)**.** Load the node table you saved in Task 2 (`nodes.csv.gz` for exp 1, site 1).

Step 1. The current table already contains `neighbor_jump_now` (which is $E_k(t)^{(0)}$). You need to construct $E_k(t)^{(\tau)}$ for $\tau = 1, 2, 3, 4, 5$. Within each track, shift the column `neighbor_jump_now` forward by τ frames:

```
# Example in pandas (nodes_df loaded from nodes.csv.gz)
nodes_df = nodes_df.sort_values(['track_id', 'Image_Metadata_T'])
for tau in range(1, 6):
    nodes_df[f'lagged_exposure_{tau}'] = (
        nodes_df.groupby('track_id')['neighbor_jump_now']
        .shift(tau)          # shift forward by tau frames within track
        .fillna(False)
    )
```

Step 2. For each $\tau \in \{0, 1, 2, 3, 4, 5\}$, compute $\mathrm{RR}^{(\tau)}$:

```
import numpy as np
results = {}
for tau in range(6):
    col = 'neighbor_jump_now' if tau == 0 else f'lagged_exposure_{tau}'
    exposed   = nodes_df[nodes_df[col] == True]['future_self_jump']
    unexposed = nodes_df[nodes_df[col] == False]['future_self_jump']
    p_exp   = exposed.mean()
    p_unexp = unexposed.mean()
    results[tau] = {'p_exp': p_exp, 'p_unexp': p_unexp,
                    'RR': p_exp / p_unexp if p_unexp > 0 else np.nan}
```

Step 3. Plot $\mathrm{RR}^{(\tau)}$ against τ (x-axis: lag in frames; optionally convert to minutes using 5 min/frame). Add a horizontal dashed line at $\mathrm{RR} = 1$ as a reference.

(a) At which lag τ^* is $\mathrm{RR}^{(\tau)}$ highest? What is the corresponding physical time $\tau^* \times 5$ min? Does this suggest a rapid or slow cell-to-cell ERK relay?
(b) Does $\mathrm{RR}^{(\tau)}$ drop below 1 at large τ, or does it plateau? What does each scenario imply about the persistence of spatial ERK coordination?
(c) Compare $\mathrm{RR}^{(0)}$ (original) with $\mathrm{RR}^{(1)}$. Is the original estimator inflated, deflated, or unaffected by the simultaneity issue for this dataset?

Limitation L2 (30 min) Non-comparable thresholds across blocks

The problem

Recall Definition 4 of the handout: when no global threshold is specified, each block b uses its own data-driven threshold $\theta_b = Q_q(\mathcal{D}_b^+)$, where $\mathcal{D}_b^+$ is the set of positive increments *within block b*.

This creates a serious comparability problem. Let $\mu_b = \mathbb{E}[\Delta S_k(t)^+]$ be the mean positive increment in block b. If a mutant cell line has systematically larger ERK pulses ($\mu_{\text{mut}} > \mu_{\text{WT}}$), its per-block threshold θ_{mut} will also be larger. The consequence:

Worked numerical example. Suppose we observe (in arbitrary ratio units):

Block	Positive increments $\mathcal{D}^+$ (sample)	$\theta_{0.90}$	% nodes called jumps
WT	$\{0.01, \ldots, 0.10\}$, mean 0.03	0.07	10%
PTEN del	$\{0.03, \ldots, 0.30\}$, mean 0.09	0.21	10%

Both blocks end up labelling exactly 10% of nodes as jumps (by construction of the quantile threshold). But PTEN-deleted cells show a *three-fold larger* typical increment. The per-block threshold hides this biological difference: we are comparing "top-10% ERK steps within WT" to "top-10% ERK steps within PTEN del", which are intrinsically different events.

Question 2. In the above example, if we instead used a single global threshold $\theta_{\text{global}} = Q_{0.90}(\mathcal{D}^+_{\text{all}})$ computed from *all blocks together*, which mutation would have more jump events? Does the relative RR estimate between mutations change? Explain why this matters for a *cross-group comparison*.

Proposed improvement: global cross-block threshold

Proposed improvement: Global threshold θ^*

Collect all positive signal increments from every block in the comparison:

$$\mathcal{D}^+_{\text{all}} = \bigcup_{b \in \mathcal{B}} \mathcal{D}^+_b = \bigcup_{b \in \mathcal{B}} \{\Delta S_k(t) \mid (k,t) \in V_b,\ \Delta S_k(t) > 0\}.$$

Define the **global threshold**

$$\theta^* = Q_q(\mathcal{D}^+_{\text{all}}).$$

Re-run the single-block analysis for each block with this fixed θ^* (using the `-jump-threshold` flag instead of `-jump-quantile`). Under the global threshold, the same increment size counts as a jump in every block, making cross-group RR values directly comparable.

Task 9 (Implement global-threshold comparison)**.** You will need to do this in two steps.

Step 1: Estimate θ^*. For each mutation's site(s), load the node table(s) and pool the positive increments. (If you only have WT data from Task 1, use those.)

```
import pandas as pd, numpy as np

all_deltas = []
for node_file in your_node_files:          # list of nodes.csv.gz paths
    df = pd.read_csv(node_file)
    pos = df['signal_delta'].dropna()
    all_deltas.append(pos[pos > 0].values)

all_deltas = np.concatenate(all_deltas)
theta_star = float(np.quantile(all_deltas, 0.90))
print(f"Global threshold theta* = {theta_star:.4f}")
```

Step 2: Re-run with global threshold.

```
python scripts/spatiotemporal_signal_propagation.py \
    --exp-id 1 --site-id 9 \
    --signal-col ERKKTR_ratio \
    --jump-threshold <theta_star> \
    --output-dir analysis_outputs_global
```

(a) Compare the fraction of nodes labelled as jumps between WT (site 1) and PIK3CA E545K (site 9) when using (i) per-block thresholds vs. (ii) the global threshold θ^*. Do the two conditions differ in jump frequency under the global threshold?
(b) Compare $\mathrm{RR}_{\mathrm{WT}}$ and $\mathrm{RR}_{\mathrm{PIK3CA}}$ under per-block vs. global thresholds. Does the ranking of conditions change? Which comparison do you trust more for a biological claim about "whose cells are more spatially coordinated"?
(c) Formally, define the *bias* of the per-block estimator. Let $\mathrm{RR}_b^{\mathrm{per}}$ be computed with θ_b and $\mathrm{RR}_b^{\mathrm{global}}$ be computed with θ^*. What conditions on θ_b vs. θ^* would make $\mathrm{RR}_b^{\mathrm{per}} > \mathrm{RR}_b^{\mathrm{global}}$? (Hint: think about which nodes shift from "not a jump" to "a jump" when the threshold is lowered.)

Limitation L3 (35 min) Binary exposure discards neighbourhood-intensity information

The problem

The exposure indicator

$$E_k(t) = \mathbf{1}[m(k,t) > 0]$$

is binary: a node is "exposed" even if only *one* of its $n(k,t)$ neighbours is jumping, and it receives the same label whether 1 or 8 neighbours are jumping simultaneously.

This collapses the entire neighbourhood into a single bit and discards two sources of potentially important information:

1. **Fraction of jumping neighbours.** A cell surrounded by 8 jumping neighbours might be more strongly stimulated than one with only 1 jumping neighbour.
2. **Intensity of the neighbourhood signal.** The mean signal $\bar{S}(k,t)$ of the neighbourhood (already computed in the node table as `neighbor_mean_signal`) carries information about the ambient signalling environment that is ignored by the binary indicator.

Question 3. Consider two nodes (k,t) and (k',t'), both with $n = 8$ spatial neighbours. Cell k has $m = 1$ jumping neighbour; cell k' has $m = 8$ jumping neighbours. Under the current formulation, both are "exposed" equally ($E_k(t) = E_{k'}(t') = 1$).

Does it seem reasonable to you that both cells should have the same expected future-jump probability p_{exp}? What biological mechanism would justify equal treatment? What mechanism would predict that k' (with 8 jumping neighbours) is more likely to jump in the near future?

Proposed improvement A: fractional exposure

Proposed improvement: Fractional exposure $\tilde{E}_k(t)$

Replace the binary indicator with the **fraction of jumping neighbours**:

$$\tilde{E}_k(t) \;=\; \begin{cases} m(k,t)/n(k,t) & \text{if } n(k,t) > 0, \\ 0 & \text{if } n(k,t) = 0. \end{cases}$$

$\tilde{E}_k(t) \in [0,1]$. Partition nodes into exposure *bins* $[0, \frac{1}{B}), [\frac{1}{B}, \frac{2}{B}), \ldots, [\frac{B-1}{B}, 1]$ and compute a separate future-jump rate p_b for each bin. A dose–response relationship (p_b increasing with bin index) would support the hypothesis that more jumping neighbours produce stronger induction.

Proposed improvement B: signal-level exposure

Proposed improvement: Neighbourhood-signal exposure $\hat{E}_k(t)$

Use the **mean neighbour signal** as a continuous exposure variable:

$$\hat{E}_k(t) \;=\; \bar{S}(k,t) \;=\; \frac{1}{n(k,t)} \sum_{(k',t) \in \mathcal{N}_k(t)} S_{k'}(t).$$

Discretize $\hat{E}_k(t)$ into quartiles Q_1–Q_4 across all nodes with $n(k,t) > 0$. Compute the future-jump rate p_{Q_j} for each quartile. An increasing pattern (higher ambient neighbourhood signal $\Rightarrow$ higher future-jump probability) would indicate that the mean state of the neighbourhood, not just its "jumpiness", predicts self-activation.

Task 10 (Implement dose–response analysis)**.** Using the node table from Task 2, implement both improvements.

Step 1: Fractional exposure.

```
df = pd.read_csv('nodes.csv.gz')

# fractional exposure: use neighbour_jump_count / neighbour_count
df['frac_exposure'] = (
    df['neighbor_jump_count'] / df['neighbor_count'].replace(0, np.nan)
).fillna(0.0)

# bin into 5 levels: 0, (0,0.25], (0.25,0.5], (0.5,0.75], (0.75,1]
bins   = [-0.001, 0.0, 0.25, 0.50, 0.75, 1.01]
labels = ['0', '(0,0.25]', '(0.25,0.5]', '(0.5,0.75]', '(0.75,1]']
df['frac_bin'] = pd.cut(df['frac_exposure'], bins=bins, labels=labels)

dose_response = (df.groupby('frac_bin', observed=True)['future_self_jump']
                   .agg(['mean', 'count'])
                   .rename(columns={'mean': 'future_jump_rate',
                                    'count': 'n_nodes'}))
print(dose_response)
```

Step 2: Neighbourhood-signal exposure.

```
has_nb = df['neighbor_count'] > 0
df.loc[has_nb, 'nb_signal_quartile'] = pd.qcut(
    df.loc[has_nb, 'neighbor_mean_signal'], q=4,
    labels=['Q1','Q2','Q3','Q4']
)
nb_dose = (df.groupby('nb_signal_quartile', observed=True)['future_self_jump']
             .agg(['mean','count'])
             .rename(columns={'mean':'future_jump_rate','count':'n_nodes'}))
print(nb_dose)
```

(a) Plot the dose–response curves for both improvements (bar chart or line plot: x-axis = bin/quartile, y-axis = p_b). Does the fractional-exposure curve ($\tilde{E}$) show a monotone dose–response? What would a flat curve (all bins equal) imply biologically?

(b) Now compute a single summary statistic analogous to RR for the fractional exposure. Define the *graded relative risk*:

$$\widetilde{\mathrm{RR}} = \frac{p_B}{p_0},$$

where p_B is the future-jump rate in the highest fraction bin and p_0 is the rate in the zero-exposure bin ($\tilde{E} = 0$). Compare $\widetilde{\mathrm{RR}}$ with the original RR from Task 1. Which estimator is larger? Why does $\widetilde{\mathrm{RR}}$ represent a "cleaner" signal?

(c) Does the neighbourhood signal level $\hat{E}_k(t)$ independently predict future self-jumping (beyond the binary jump indicator)? In other words, do cells in high-signal neighbourhoods (Q4) jump more in the future even when $E_k(t) = 0$ (no jumping neighbour)? Filter the data to `neighbor_jump_now` = False and repeat the quartile analysis for this sub-population.

Synthesis (bonus, 10 min)

Task 11 (Combining all three improvements)**.** You now have three independently derived improvements:

Limitation	Improvement	Key new quantity
L1: simultaneity	Lagged exposure	$E_k(t)^{(\tau)}$, $\mathrm{RR}^{(\tau)}$
L2: non-comparable thresholds	Global threshold	θ^*, $\mathrm{RR}_b^{\mathrm{global}}$
L3: binary exposure	Fractional/signal exposure	$\tilde{E}_k(t)$, $\widetilde{\mathrm{RR}}$

Suppose you wanted to combine improvements L1 and L3 into a single estimator. Write down the mathematical definition of the *lagged fractional exposure*

$$\tilde{E}_k^{(\tau)}(t) = \frac{m(k,\, t-\tau)}{n(k,\, t-\tau)},$$

and the corresponding graded relative risk

$$\widetilde{\mathrm{RR}}^{(\tau)} = \frac{p_B^{(\tau)}}{p_0^{(\tau)}},$$

where $p_B^{(\tau)}$ is the future-jump rate for nodes in the highest fractional-exposure bin *at lag* τ.

(a) Implement $\widetilde{\mathrm{RR}}^{(\tau)}$ for $\tau \in \{0, 1, 2, 3, 4\}$ and plot the resulting lag curve. Does it peak at the same τ^* as the binary $\mathrm{RR}^{(\tau)}$ from Task 8?
(b) Open discussion (no single correct answer): What further mathematical improvement would you propose? Some options to consider:

- Accounting for the directionality of cell movement (use a directional neighbourhood cone rather than a symmetric ball).
- Using a continuous signal increment $\Delta S_k(t)$ instead of the binary $\mathcal{J}_k(t)$ to define a "soft jump" probability via a sigmoid function.
- Replacing the empirical mean estimators for p_{exp} and p_{unexp} with a mixed-effects logistic regression that treats the track as a random effect, properly handling the non-independence of nodes.

Describe your chosen improvement in mathematical notation and sketch the implementation.

Notation follows the companion handout exactly. All tasks assume Python 3, pandas, NumPy, and the project virtual environment. Refer to the notation table in the Appendix of the handout for symbol definitions.